

SCRIBE: Consistent Distributed Environment Modeling for Communication-Limited Robotic Exploration

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Abstract—SCRIBE addresses distributed environment-model learning under limited communication while preserving accurate, mutually consistent estimates of the field and its uncertainty. It represents spatial models as fixed-dimensional linear states and introduces a fusion process that lets agents repeatedly assimilate any available peer’s model without full-team participation. Information Kalman filtering adds new evidence, Covariance Intersection reconciles divergent priors, and Metropolis–Hastings consensus propagates innovations using only local neighbor information. The resulting model update is lightweight, asynchronous, and consistent under intermittent communication. In 96 missions with 3–10 agents, final normalized error is 0.405–0.409, against 0.582–1.227 without communication. Across 432 planning trials, median fusion through ten agents remains below 6.2 ms and 24 kB per update. Independent Vulcan planners achieve pooled posterior variance within 3.2% of ideal centralized sharing, demonstrating effective exploration through communication loss.

Index Terms—autonomous exploration, distributed estimation, informative path planning, intermittent communication, multi-robot systems

I. INTRODUCTION

Autonomous exploration uses an uncertain environment model to decide which observations are most valuable [1], [2]. This is particularly important in the deep ocean, where vehicles may localize plumes, characterize cold seeps, or follow chemical gradients from sparse data [3]. Teams expand coverage, but acoustic links are intermittent, bandwidth-limited, and range-dependent [1]. Robots must therefore learn and plan in isolation, then incorporate peer knowledge whenever contact resumes.

Distributing environment-model learning is non-trivial: each robot requires a locally useful model, yet team models must remain consistent in prediction and confidence. Intermittent links cause robots to accumulate different evidence and their learned states to diverge. Existing methods either exchange observations and relearn from the collective history, whose cost grows with mission duration and sensor dimension [4], [5], or exchange and average compact model parameters, which can count correlated information more than once and overstate certainty [6]. The fundamental problem is not communication reduction alone, but guaranteeing an accurate, consistently calibrated shared model when only subsets of the team can participate in each update.

Distributed GP updates [7], finite-dimensional regression [8], and model-based planning [9], [10], [11] establish the value of collaborative learning. SCRIBE contributes a model-learning and fusion mechanism that remains consistent under intermittent communication. A shared GSF or EOF basis first recasts the spatial model as a fixed-dimensional linear

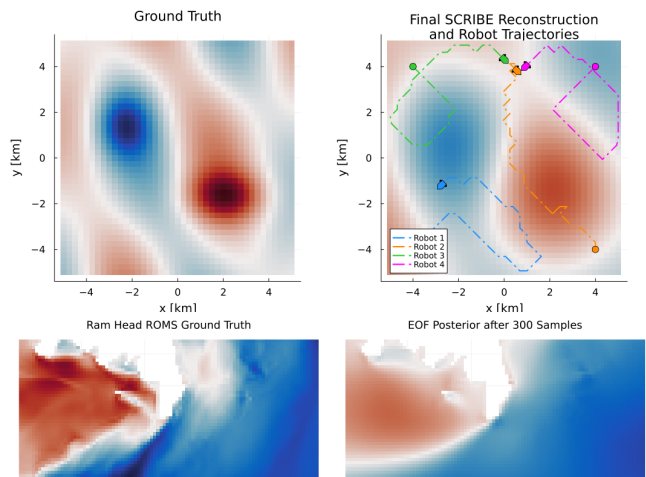


Fig. 1. Two SCRIBE models. Top: a simulated field, one robot’s final GSF estimate, and four robot paths. Bottom: held-out Ram Head ROMS snapshot 5021 and its EOF estimate after 300 samples ($b = 2$, RMSE 0.049).

coefficient state. SCRIBE then interleaves an Information Kalman Filter, whose observation updates are additive, with distributed Covariance Intersection, which reconciles divergent priors without their unknown cross-correlations [6], [12]. Metropolis–Hastings consensus propagates the new information using only current neighbor degrees [13]. This hybrid construction [14] allows any available agents to repeatedly fuse their learned fields without loss of consistency, a global graph, or a synchronized team-wide update.

II. DISTRIBUTED ENVIRONMENT MODELING

A. Fixed-Dimensional Information State

When no collection of earlier field maps is available, SCRIBE represents a scalar field with n_ϕ Gaussian scalar-field (GSF) basis functions [8], [15],

$$\mathcal{F}(x, k) = \psi^T(x) \phi(k), \quad \psi_i(x) = \tau_i^{-1} e^{-\|x - \mu_i\|_2^2 / \sigma_i^2}. \quad (1)$$

The centers μ_i and widths σ_i place overlapping functions across the region; the coefficients $\phi_i(k)$ determine how strongly each function appears in the field at time k .

When a prior environmental record is available, a mean-centered, spatially weighted singular value decomposition instead yields empirical orthogonal function (EOF) modes [16]. If $\mathbf{E}_r$ contains the r retained modes at the stored feature locations, the field vector is

$$\mathbf{f}(k) = \bar{\mathbf{f}} + \mathbf{E}_r \phi(k), \quad (2)$$

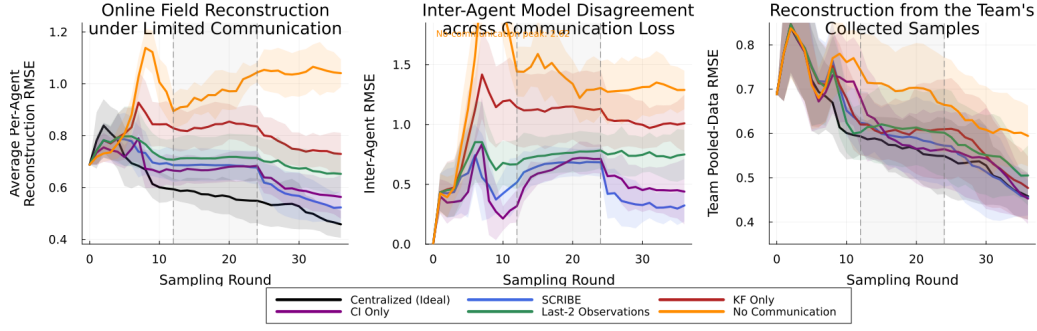


Fig. 2. Twelve four-robot trials of 36 rounds. Shading marks complete blackout; otherwise each robot communicates with at most one in-range peer. Curves give the mean and one standard deviation. The right panel rebuilds a model from all samples chosen by each team and therefore compares path quality rather than model fusion. The disagreement axis is truncated; lower is better.

where the prior mean $\bar{\mathbf{f}}$ and modes remain fixed while $\phi(k)$ describes the current field. GSFs suit a poorly known environment, while EOFs compress spatial structure present in earlier data.

Online estimation differs only in constructing the observation matrix. GSF rows are $\psi^T(x)$ at the sample locations. For EOFs, interpolation $\mathbf{L}_{\mathbf{X}_j}$ maps the stored grid to those locations, giving $\mathbf{H}_j = \mathbf{L}_{\mathbf{X}_j} \mathbf{E}_r$ and centered data $\mathbf{z}'_j = \mathbf{z}_j - \mathbf{L}_{\mathbf{X}_j} \bar{\mathbf{f}}$. GSF data require no mean subtraction. Both satisfy $\mathbf{z}'_j = \mathbf{H}_j \phi + \mathbf{v}_j$. Agent j stores $\mathbf{Y}_j = \mathbf{P}_j^{-1}$ and $\mathbf{y}_j = \mathbf{Y}_j \phi_j$; a measurement adds

$$\delta \mathbf{I}_j = \mathbf{H}_j^T \mathbf{R}_j^{-1} \mathbf{H}_j, \quad \delta \mathbf{i}_j = \mathbf{H}_j^T \mathbf{R}_j^{-1} \mathbf{z}'_j. \quad (3)$$

SCRIBE exchanges an $O(n_\phi^2)$ state independent of mission duration, rather than an $O(kd_z)$ observation history. From $\hat{\phi}_j = \mathbf{Y}_j^{-1} \mathbf{y}_j$, GSFs predict $\psi^T(x) \hat{\phi}_j$; EOFs use $\mathbf{L}_{\mathbf{X}}(\bar{\mathbf{f}} + \mathbf{E}_r \hat{\phi}_j)$.

B. Fusion under an Unstable Network

Agents' priors may contain shared observations, so adding them can count the same evidence repeatedly. Let $\mathcal{N}_j^{\ell+} = \mathcal{N}_j^\ell \cup \{j\}$ and $\Delta_j^\ell = \{\omega \geq 0 : \sum_{m \in \mathcal{N}_j^{\ell+}} \omega_m = 1\}$. Distributed CI fuses these correlated priors without requiring their cross-correlations:

$$\omega^* = \arg \min_{\omega \in \Delta_j^\ell} J \left[\left(\sum_{m \in \mathcal{N}_j^{\ell+}} \omega_m \mathbf{Y}_m^\ell \right)^{-1} \right], \quad (4)$$

$$(\tilde{\mathbf{y}}_j^{\ell+1}, \tilde{\mathbf{Y}}_j^{\ell+1}) = \sum_{m \in \mathcal{N}_j^{\ell+}} \omega_m^* (\mathbf{y}_m^\ell, \mathbf{Y}_m^\ell). \quad (5)$$

Current measurement innovations are new evidence and are instead averaged by MHMC. For $\delta \mathbf{s}_j \in \{\delta \mathbf{i}_j, \delta \mathbf{I}_j\}$,

$$\gamma_{jm}^\ell = \frac{1}{1 + \max(|\mathcal{N}_j^\ell|, |\mathcal{N}_m^\ell|)}, \quad (6)$$

$$\delta \mathbf{s}_j^{\ell+1} = \delta \mathbf{s}_j^\ell + \sum_{m \in \mathcal{N}_j^\ell} \gamma_{jm}^\ell (\delta \mathbf{s}_m^\ell - \delta \mathbf{s}_j^\ell). \quad (7)$$

The weights require only current neighbor degrees. After consensus within a component of size n_c ,

$$\mathbf{y}_j^+ = \tilde{\mathbf{y}}_j + n_c \delta \mathbf{i}_j, \quad \mathbf{Y}_j^+ = \tilde{\mathbf{Y}}_j + n_c \delta \mathbf{I}_j. \quad (8)$$

CI therefore preserves consistent confidence while MHMC distributes new observations. Fusion begins whenever a link appears, without a global graph or common schedule.

Computational scaling. For basis size b , messages are $\Theta(b^2)$ and assimilating q_j samples is $O(q_j b^2)$. An MHMC round is $O(d_j b^2)$ locally and $O(E b^2)$ fleet-wide; CI is $O(J_{\text{CI}}(d_j b^2 + b^3))$. Hence bounded-degree fleets scale as $O(N b^2)$ per round, versus $O(N^2 b^2)$ for complete graphs. Fixed-graph convergence is $O(\log(e_0/\epsilon)/(1-\rho))$; for changing links, it depends on the link process.

III. EXPERIMENTS AND RESULTS

Consistency after reconnection. Four agents first follow matched paths, isolating fusion from planning. Across 30 trials with disjoint links, a middle-third blackout, and reconnection, SCRIBE reaches RMSE 0.117 ± 0.011 , close to centralized sharing (0.101 ± 0.012) and CI (0.118 ± 0.012), but below IKF without CI (0.288 ± 0.020) and no communication (0.814 ± 0.018). Reconnection therefore does not treat correlated priors as independent evidence. Fig. 1 also shows that EOFs trained on prior Ram Head ROMS output [17] retain the dominant held-out structure; the two-mode estimate reaches RMSE 0.049 after 300 samples.

Fusion without scheduled communication. Agents then sense continuously for 8178 steps in adjacent regions and exchange models only at proximity encounters. Every team size in Fig. 3 converges near centralized NRMSE (0.404–0.409), versus 0.582–1.227 without communication. The six-agent trajectories converge from distinct initial errors; residual spatial smoothing is the two-mode representation floor, not an asynchronous-fusion failure.

Operational scaling. Across 432 trials with 2–10 agents, Fig. 4 separates estimation quality from fusion cost. Posterior variance remains stable, while network mixing governs consensus rounds, time, and payload. The worst case, the intermittent moderate-degree graph at ten agents, requires about 26 rounds, 6.2 ms, and 24 kB per update. Model

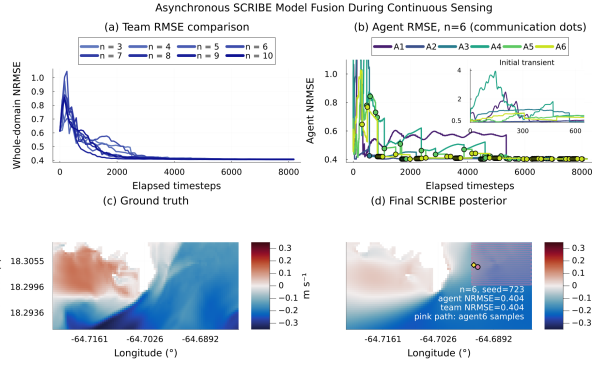


Fig. 3. Asynchronous fusion during continuous sensing. Circles mark peer encounters in the selected six-agent mission; the bottom row compares its final model with the ROMS ground truth.

quality therefore persists as the fleet grows; connectivity determines the operational cost.

Effect on information-guided exploration. Four independent Vulcan planners [18] explore for 36 rounds with distance-limited links and a twelve-round blackout in Fig. 2. SCRIBE has the lowest final distributed error and, like CI, contracts disagreement after reconnection; IKF remains high because it does not reconcile correlated priors. SCRIBE’s final RMSE (0.523 ± 0.060) approaches centralized sharing (0.458 ± 0.052), improves on CI (0.564 ± 0.084), and halves the no-communication error (1.041 ± 0.053).

Pooling each regime’s samples isolates path quality. SCRIBE nearly matches centralized pooled RMSE (0.455 versus 0.458) and integrated variance (0.0488 versus 0.0473), the latter within 3.2%. Consistent fusion thus recovers almost all information gathered under ideal sharing, without joint planning or team-wide communication.

IV. CONCLUSION

SCRIBE provides consistent, fixed-dimensional fusion through intermittent communication. Tests through ten agents approach centralized information collection, while $O(b^2)$ messages keep local models usable during blackouts. The planner-independent update supports distributed GP and MCTS systems [9] or Multi-Agent Vulcan [19] across remote missions.

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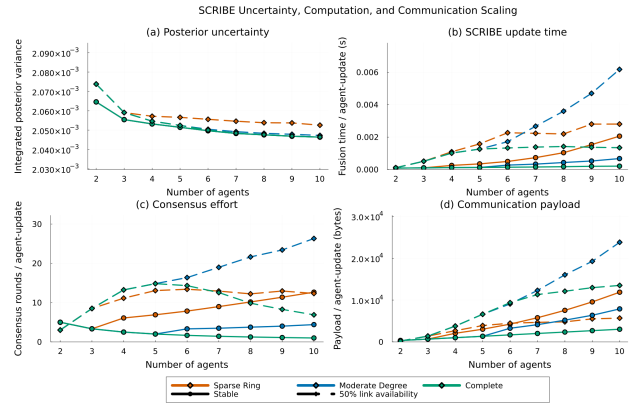


Fig. 4. Uncertainty, computation, and communication scaling across team sizes, network densities, and link conditions. Dashed curves denote 50% link availability.

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